A Comparison of Lexical and Contextual Embedding Representations for Job Posting Analysis

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Abstract

The increasing availability of online job insertions makes the automatic analysis a useful tools to support hiring processes and research. However, descriptions are usually non structured, heterogeneous and characterized with variable semantic. In this project two main task are developed: classification and topic modeling. Different textual representation are compared based on both lexical statistics and contextual embeddings. Linear and non-linear models are used to classify and probabilistic and neural methods for topic modeling. Results show that, for classification, TD-IDF with random forest achieves better performances, showing that specific words in the domain are in this case more discriminant compared to deep semantic representation. In topic modeling a compromise between coherence and diversity is needed. LDA offers more balanced results while models based on embedding offer less interpretable topics. Overall, the analysis shows that simple methods can still be efficient in daily applications.

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1 Introduction

The rising number of online jobs insertions makes the analysis of the labor market an interesting source of information, weather your are looking for one or seeking a new employee. Job's descriptions usually include required competences, responsibilities, and professional conditions. The availability of textual data, even if descriptions are usually unstructured and heterogeneous, is rich of semantic information. We believe that extracting and organizing this type of data is an interesting challenge that could help and automatize job's searches.

In this project, we focus on two related tasks: classification of job insertions based on the specific position, and the research of latent topics in the jobs' descriptions.

- **Text classification**: The aim is to build a model capable of mapping a description back to its profession title;
- **Topic modeling**: the aim is to detect coherent groups that share similar semantic contents.

This analysis includes both supervised and unsupervised methods.

The original dataset is composed by 30.570 ads and 12 columns, but for our work we needed only two: the job title and the job description. The dataset, after removing duplicates, includes 615 different job positions with slightly unbalanced classes. This context reflects realistic conditions and justifies the necessity of models able to deal with textual and high-dimensional data, with strong semantic variability among the job descriptions.

2 Preprocessing

In this project a dual-pipeline strategy is applied for preprocessing, distinguishing between statistical approaches (Topic Modeling) and neural approaches (Embeddings), consistently with the most efficient practices in text representation.

2.1 Job labels

A crucial point in the pipeline was the removal of jobs' titles from the descriptions texts. Without this step, the model would trivially learn to recognize the label string rather than the description's semantic content, leading to inflated and unrealistic performance metrics. The plot 3 in Appendix shows TF-IDF and BERTino's distributions of the jobs change when we keep ("With Leakage prevention") and remove ("Without Leakage prevention") labels. We can see that there is not a clear separation among the points in none of the plots, and this will inevitably impact the accuracy in the text classification.

2.2 Basic preprocessing

Aggressive preprocessing was applied to topic modeling. For tasks relying on Bag-of-Words (BoW) models like LDA and LSA, we applied aggressive cleaning, including the removal of stop words (e.g., "il", "per", "azienda") and non-alphabetic characters. This was necessary because BoW models rely entirely on term frequency. Words that carry little semantic content could otherwise dominate probability distributions, obscuring meaningful topics.

2.3 Embeddings

Conversely, for the multi-sentence-BERTino model, a minimal preprocessing has been employed (casing preservation, retaining stopwords). Contextual embedding models generate vector representations based on the entire sequence of words; removing stopwords or stemming would destroy the syntactic context (e.g., "to" in "to be") required by the model to resolve ambiguity and polysemy.

3 Text Representation

The choice of the textual representation plays a core role in determining the effectiveness of both classification models and topic modeling techniques. Different representations codify linguistic information at different levels, from superficial lexical statistics to deeper seman-

tic and contextual relations. In this work, two complementary representations are considered: one sparse representation based on term frequency, and one based on embeddings. These choices were made to analyze how the structure of the representational space influences the performances of the models, their interpretability, and robustness.

Sparse representations, such as TD-IDF, model the documents as vectors in a high dimensional space, where each dimension is a term of the vocabulary. These types of representation are easily interpretable and can be adapted to linear models, and probabilistic topic modeling techniques, but they are limited when semantic similarity is the target. On the other hand, dense representations, obtained by transformer models, project the documents in a lower dimensional space, where semantic relations are implicitly codified. Although richer, these methods introduce more complexity in geometry and interpretation.

3.1 TF-IDF

As mentioned above, for this representation stop words were removed. This approach effectively highlighted discriminative keywords (e.g., "java" for developers, "saldatura" for welders). However, it suffers from the "vocabulary mismatch" problem, failing to capture synonyms or semantic relationships (e.g., understanding that "coder" and "developer" are related). From a geometric perspective, TF-IDF represents documents as sparse vectors lying in a very high-dimensional space. The similarity between documents is driven primarily by shared vocabulary, making this representation particularly effective when job positions are characterized by domain-specific keywords. For this reason, TF-IDF serves both as a strong baseline for classification and as a natural input for linear topic models such as LSA and probabilistic models such as LDA and HDP.

3.2 BERTino

BERTino[1] is a pre-trained model useful for dealing with texts in Italian. By mapping texts to 768-dimensional vectors, this approach captures deep semantic relationships and context. The significant performance gap between classifiers using these embeddings suggests that, while the embeddings contain rich semantic data, the spatial distribution of these classes is complex and non-linear. Unlike TF-IDF, BERTino embeddings encode contextual meaning and semantic relationships between words, allowing documents with different vocabularies but similar intent to be placed close to each other in the embedding space. However, this increased expressiveness results in a complex, non-linearly separable geometry, which explains the performance gap observed between linear and non-linear classifiers. Although this representation is less interpretable at the token level, it is particularly suitable for capturing higher-level semantic similarity and supporting clustering-based topic modeling approaches.

4 Text Classification

The classification task is challenging because of the high number of job positions and the semantic overlap between the job descriptions. Many professions share similar tasks and terminology, making it difficult for the model to outline a substantial difference.

4.1 Classification Models

At this point, the description are ready to be classified. Three different models are employed for both approaches: logistic regression, random forest, and stochastic gradient descent.

4.1.1 Model evaluations

The table below suggest that, even if the lexical representation of the description seems superficial, they are informative, and a non-linear classifier as random forest can make better use of these information with respect to linear models. This result is common when the classes are characterized by recurrent key words and superficial combinations of terms. TD-IDF with Logistic Regression achieves an accuracy of 0.522. This performance suggests that linear separability between classes exists, but it is limited: the signal is present but it is not simple enough to be captured by a linear model. BERTino models do not perform well, why? Among them, BERTino with Random Forest has higher accuracy(0.5864) but it still far from the best model TF-IDF with Random Forest. This suggests that contextual embeddings do not add enough discriminant information that is valuable for this specific task, or that they are not exploited enough by the classifier. Let's take a look at BERTino with Logistic Regression (0.3448) and with SGD (0.2562). Such low accuracies show how BERTino's representations are not separable by linear models with respect to the labels, at least not in our environment. Overall, the results' table shows that in this framework simple models such as TD-IDF works better than pre-trained models. The specific lexicon of the domain seems to be more important than the semantic meaning, and Random Forest is the best one to take advantage of this particular situation.

Evaluation metrics

Model	Accuracy
TF-IDF + LogReg	0.5226
TF-IDF + RF	0.6618
TF-IDF + SGD	0.4850
BERTino + RF	0.5864
BERTino + LogReg	0.3448
BERTino + SGD	0.2562

4.2 Error analysis

The most common mistakes involve the term "tecnico". The models are not able to distinguish between a technician or a more general operator in the same field. Descriptions are assumed to be too broad without explicitly stating that more advanced skills are required. One of the most confused pair is private driver vs hearse driver. The description doesn't have enough context to distinguish between what will be driven. Similarly, with "montatore aeronautico" and "montatore di motocicli". Mistakes between "ingegnere ambientale" and "consulente ecologico" implies that description share common words as "environment" but not specifications about the degree of the profession. Nevertheless, these two figures could overlap in real life situations. Another mistake is found in same job position with different job contracts, with one being a freelancer and one being a State employee. More generally, it seems like some descriptions don't have enough details to distinguish between jobs in the same filed, but with slightly different responsibilities or between position with the same job applied on different objects.

5 Topic Modeling

Topic modeling is the second task chosen for the project. The aim is to have a topic representation where jobs labels are grouped by the similarity of their tasks.

Topic quality is evaluated using coherence, diversity and perplexity when reasonable. The metrics capture complementary aspects of the models, semantic consistency and the ability to distinguish between different thematic ar-

eas, and fitting on the data.

5.1 Data preparation

The dataset has been divided in train (70%) and test (30%). HDP has been used to choose the optimal number of topics, 18, and it has been given to the other models. BERTino has also a specific function to find the optimal number of topics, but to be coherent, the output number of HDP has been kept. The following table contains the models and their characteristics:

Model	Type	Key Characteristics
LSA	Matrix factorization	SVD on TF-IDF matrix
LDA	Probabilistic	Dirichlet priors, interpretable
HDP	Non-parametric	Automatic topic number
BERTopic	Neural	Embeddings + clustering

5.2 Results

The models are assessed through Coherence, Perplexity and Diversity. Coherence measures weather the words in each group are semantically correlated. Higher values indicate that the topics are more interpretable by a human. Perplexity measures if a probabilistic model fits the data, and in this case it can only be applied to LDA and HDP. Log-perplexity was computed for the project. Diversity measures if the topics differ from each others. Together, they provide quantitative insights on the topics.

- Hierarchical Dirichlet Process (HDP) has the highest value for coherence (0.65), here the topics are on average more interpretable. However, diversity decreases indicating a possible overlap between groups. HDP automatically chooses the number of topics, and this could lead to partially redundant groups(CHEKC);
- BERTopic has controversial metrics. Coherence is not optimal, on average, but diversity is the highest overall. This means that topic are well separated between each others but not they are not consistent. This might happen while working with embeddings, where groups are far in the semantic space but the words are not compact inside of them;
- LDA shows the best compromise. Coherence is higher than LSA and diversity is good (0.79). Perplexity has a low value meaning it can explain the corpus. LDA produces topic that are separated, meaningful and correct on the probabilistic level;
- LSA has low values for both coherence and diversity. Topics are not internally defined and similar between each others. This might be due to the algebraic nature of the model.

A critical trade-off emerged between Coherence and Diversity. LDA and HDP perform better if the aim is to explore the data and interpret the topics. On the other hand, BERTino is able to cover more content but at the expense of the clarity inside the topic. An HTML page has been generated for each model in order to explore the results with an interactive interface. The file can be found in the Google Drive folder.

The results have been summarized in the following table :

Model	Coherence	Perplexity	Diversity
LSA	0.5398	/	0.5550
LDA	0.6236	-8.5202	0.7900
HDP	0.6493	-14.0123	0.6350
BERTopic	0.5509	/	0.8421

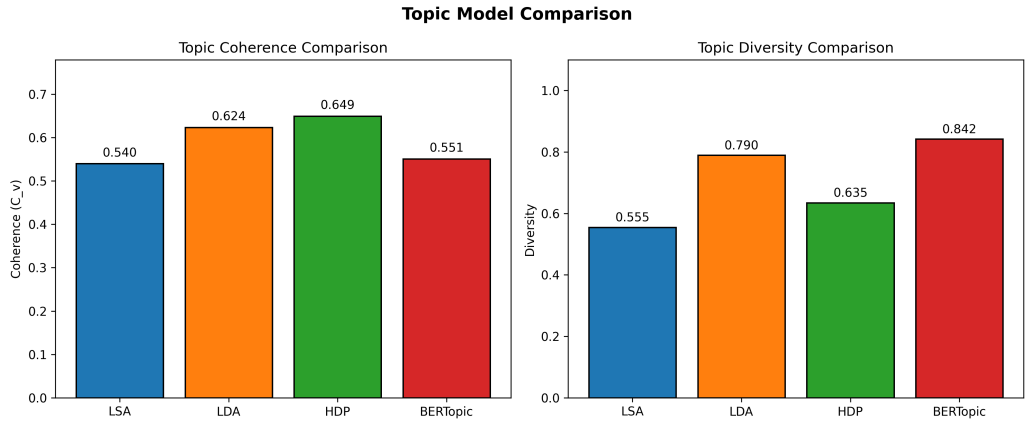

Figure 1: Coherence and Diversity

The following plot shows first eight topics generated by BERTopic, as it has more appropriate groups. Even if LDA had a more balanced score, its topic seemed farther away from the project goal.

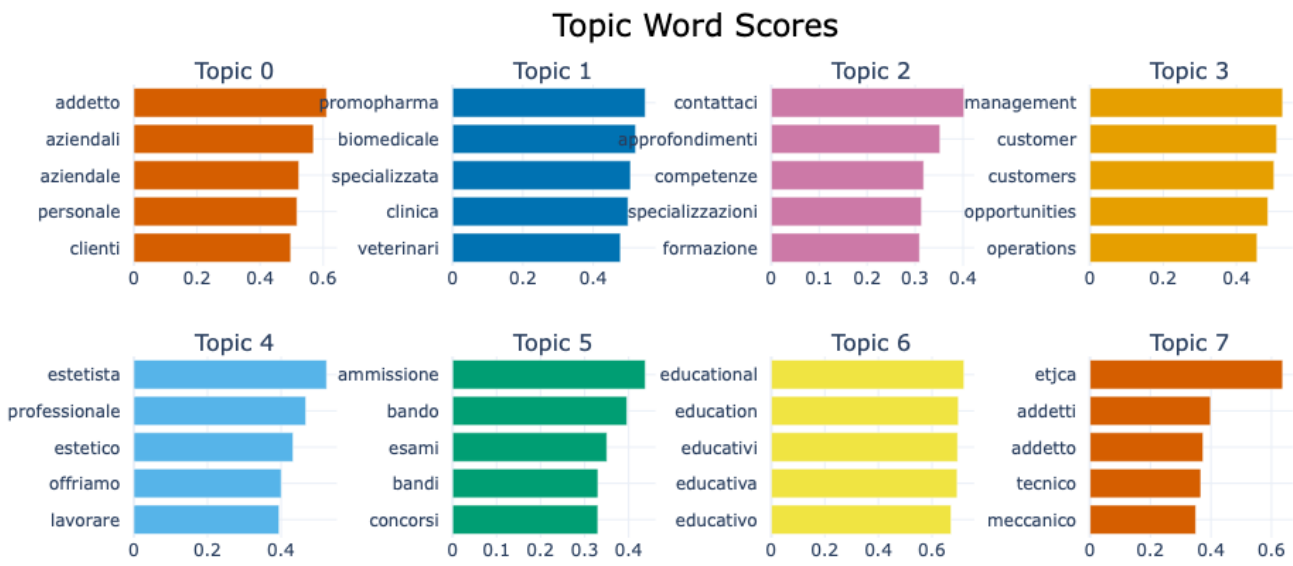

Figure 2: BERTopic topics

6 Conclusion

The analysis of job descriptions was made through two complementary tasks: supervised classification of insertions based on the profession title and the identification of topics. The evaluation showed how different textual representation influenced models, interpretability and quality on an everyday context.

Classification results show that simple approaches based on TD-IDF with non-linear classifier such as Random Forest perform better than more complex models based on embeddings. This means that, in the context of labor insertions, key recurrent words have a more important role than semantic comprehension of the entire text. BERTino's representation are mapped in a complex space where both linear and non linear models struggle. This shows that pre-trained tools doesn't necessary mean better performances, especially in absence of fine tuning and semantically separable classes.

The analysis of errors supports this with evidence: most of the confusion is caused by very general or vague descriptions, reflecting limits of the text itself rather than being only computational limits.

Topic modeling showed the trade off between internal coherence and diversity among topics. LDA stands as it balances interpretable topics with no particular overlapping, with probabilistic support. HDP reaches good results as well but is a little more redundant. Scores better in perplexity but this is probably due to the fact that the number of topic for all models was chosen by it. BERTopic sacrifies internal meaning focusing on group separation.

References

[1] Nick Prock. multi-sentence-bertino, 2023. Hugging Face model card, accessed: 2026-01-30.

A Appendix

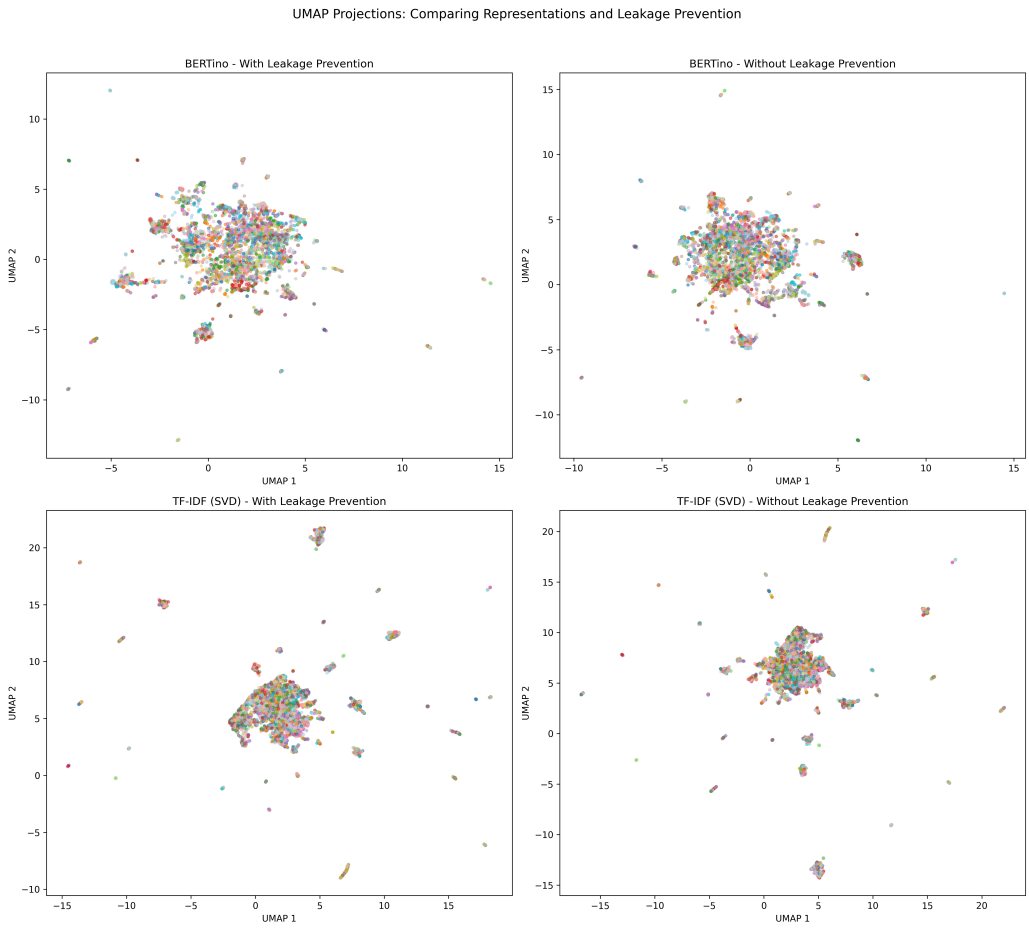

Figure 3: TD-IDF and BERTino distribution